# Project: 101 alphas in cryptocurrencies market

## Executive Summary:

This report presents the development and evaluation of quantitative trading strategies in the cryptocurrency market based on the *WorldQuant 101 Alphas* framework. A subset of alphas was selected by analyzing their cross-sectional predictive power, measured via Information Coefficient (IC) statistics, with emphasis on signals exhibiting statistically significant t-statistics ($|t| > 1$).

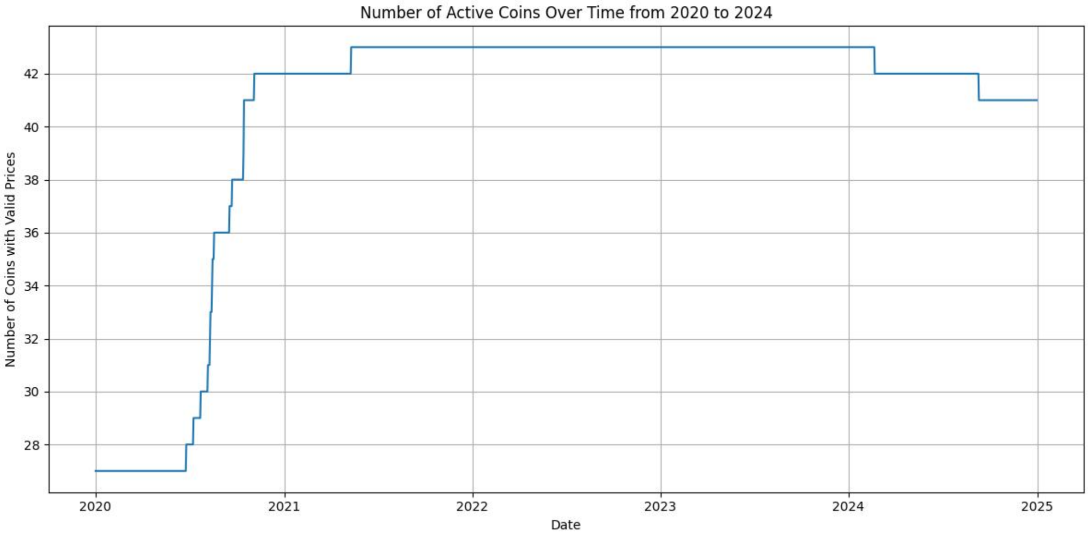
To enhance robustness, multiple weak alphas were combined into composite "super alphas" using several aggregation methods, including equal weighting, IC-based weighting, and inverse-correlation weighting. These combined signals demonstrated improved stability, with a positive IC ratio of approximately 51% and moderate turnover around 25%, indicating consistent predictive behavior with manageable trading intensity.

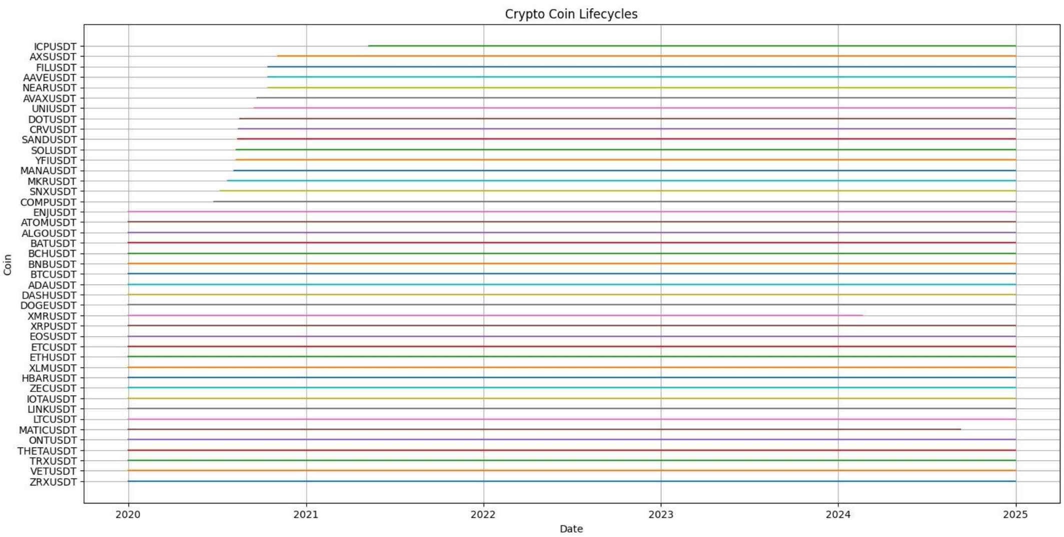
Portfolio strategies were constructed based on these super alphas under various configurations (long-short and long-only, with and without transaction costs). Backtesting results show that the strategies achieve strong risk-adjusted performance, with annualized Sharpe ratios of approximately 1.5 in-sample and 2.1 out-of-sample. These results outperform a benchmark equal-weighted long-only-all-coin portfolio across key metrics, including cumulative returns, maximum draw-down, and terminal wealth. Overall, the findings suggest that combining multiple weak signals into diversified alpha portfolios can produce robust and scalable trading strategies in the cryptocurrency market.

## Data source and data cleanness:

**Alphas**: come from the famous 101 Formulatic Alphas paper from WorldQuant (arxiv: 1601.00991). Such alphas are designed for US-stock market, with an average holding period of 0.6-6.4 days and pair-wise correlation around 15.9%.

**Pricing data:** come from the website cryptodatadownload.com/, a free and open-source platform for accessing historical cryptocurrency market data. Specifically, we select Binance as the trading venue and utilize daily OHLCV (Open, High, Low, Close, Volume) data for a broad universe of cryptocurrencies (43 coins), containing different types of coins like Layer 1, DeFi tokens and Exchange tokens, to ensure diversity. It should be noted that we include delisted assets mitigate survivor-ship bias in our backtesting. The sample period spans from 2020 to 2025, covering a full market cycle that includes both bullish and bearish phases of the cryptocurrency market, and we split it into training data (from 2020 to 2024) and test data (2025). An overview of the number of survived coins across the year and the life cycle of each coins can be found in the following graphs:




## Alpha selecting:

**A first shortlist** excludes those alphas that are not adequate for cryptocurrencies market, for example:

- those require equity-specific data like industry/sector (e.g alpha 59), since such data are unavailable in cryptocurrencies market;
- those with very long horizons (e.g alpha 79), since crypto-regmine changes fast;
- and those over complicated (e.g alpha 100)

At this stage, we shortlisted approximately 20 alphas spanning a diverse range of underlying economic intuitions:

- **mean-reversion:** #14, #20, #25, #30, #43

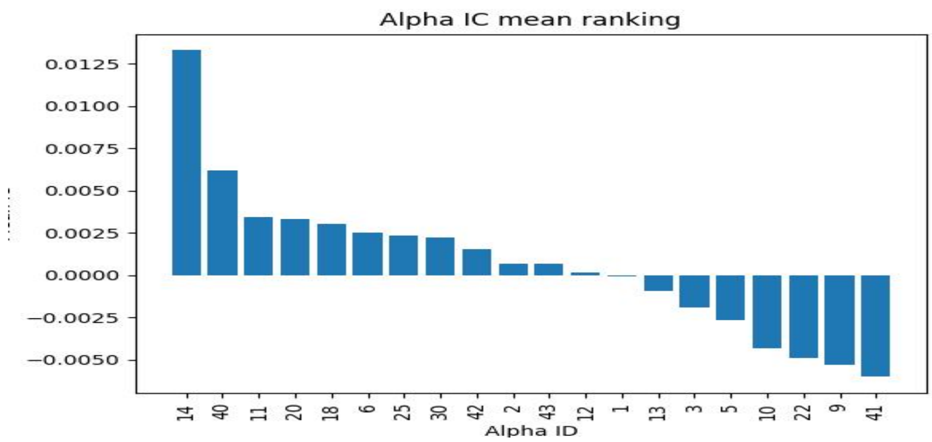
- **Momentum:** #9, #10, #12
- **Market-microstructure:** #5, #11, #41, #42
- **Volatility-based:** #1, #8, #40
- **Volume-priced interaction:** #2, #3, #6, #13, #22

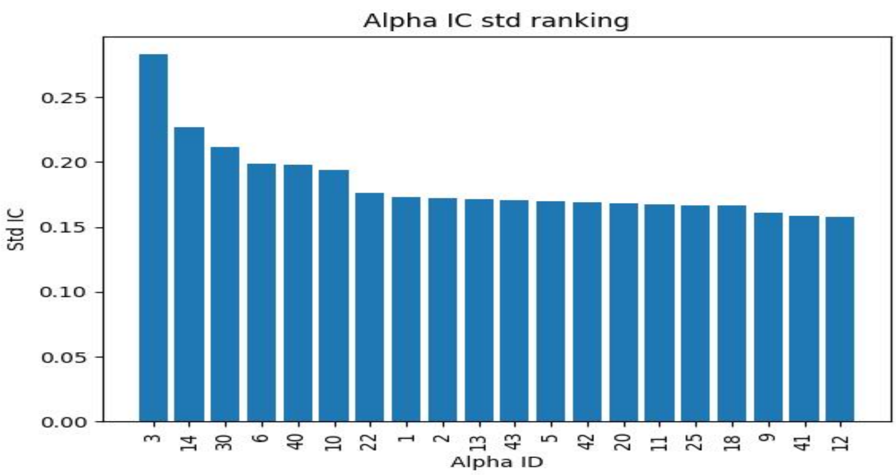
It is important to note that some alphas may capture multiple effects, for instance, Alpha #30 reflects both mean-reversion and volume–price interaction, making strict classification challenging. Nevertheless, we aim to maintain a balanced representation across categories to ensure diversification and robustness in the selected alpha pool.

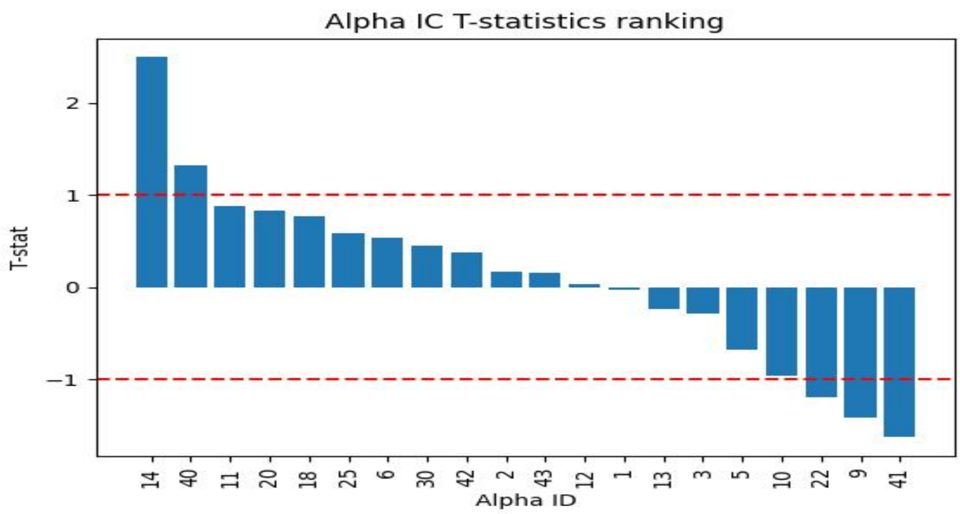
Next, we implemented these alphas and backtested them in the training dataset. More specifically, we calculated and visualized their Information coefficient (IC), the correlation between today's alpha and tomorrow's return:

$$IC_t = corr(\alpha_t, r_{t+1}).$$

To avoid look-ahead bias, we only use the information till today's closing price to calculate the alpha values. Furthermore, we plot the mean, standard deviation and t-statistics of ICs of different alphas, ranking in a descending order:






Based on the IC t-statistics, we further refine our selection by shortlisting the following alphas for subsequent analysis: **#14, #40, #22, #9, and #41**. These alphas exhibit statistical significance, with IC t-values exceeding 1, after applying sign adjustments where necessary according to their mean IC.

## Combining alphas to get super-alphas:

The motivation for constructing super-alphas stems from the observation that individually selected alphas exhibit weak out-of-sample performance. For instance, Alpha #14, which achieves the highest IC t-statistic in-sample, deteriorates to an out-of-sample IC t-statistic of only 0.44 on the test dataset. This indicates that such individual alphas lack robustness and are prone to overfitting, capturing noise rather than persistent predictive signals.

Therefore, we are motivated to combine multiple weak signals into a stronger and more stable composite signal, referred to as a *super-alpha*.

A naive approach to combining alphas is to directly sum them (after potential sign adjustments). However, this leads to a dominance effect, where alphas with larger numerical scales disproportionately influence the combined signal, resulting in limited improvement in out-of-sample performance.

A more effective approach is to first apply **rank normalization**. Specifically, for each cross-section, alpha values are transformed into ranks and re-scaled to lie within the interval $[-0.5, 0.5]$. This ensures comparability across alphas and mitigates the impact of extreme values. The normalized alphas can then be combined meaningfully.

We consider three weighting schemes to construct super-alphas:

· **Equally weighted:** Each normalized alpha is assigned the same weight. This simple approach avoids overfitting and serves as a robust baseline.

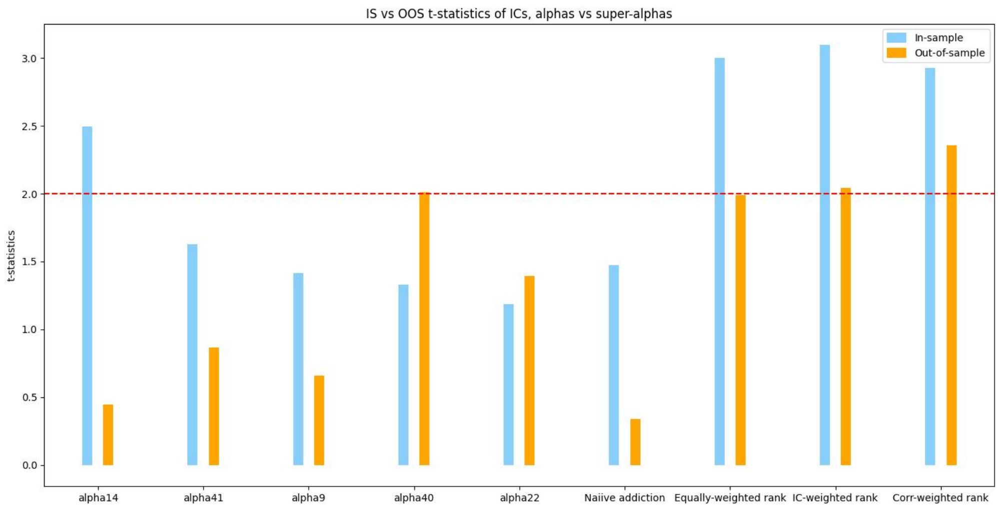
· **IC-weighted:** Weights are assigned proportionally to the in-sample IC (or IC t-statistic) of each alpha, giving more importance to signals with stronger predictive power.

· **Correlation-inverted weighted (with regularization):** Weights are determined by accounting for the correlation structure among alphas, assigning higher weights to less correlated signals to enhance diversification. Regularization is introduced to stabilize the covariance (or correlation) matrix, which can otherwise be ill-conditioned or noisy due to limited sample size and high dimensionality. This prevents extreme or unstable weight allocations and improves out-of-sample robustness.

We use the following three metrics to evaluate the quality of our super-alphas:

**1. In-sample and out-of-sample IC t-statistics.**

These metrics assess both the predictive power and statistical significance of the signals. As illustrated in the figure below, there is a clear improvement when moving from individual meta-alphas (including the naive sum) to rank-normalized super-alphas, particularly in out-of-sample performance. This highlights the effectiveness of normalization and diversification in enhancing robustness.



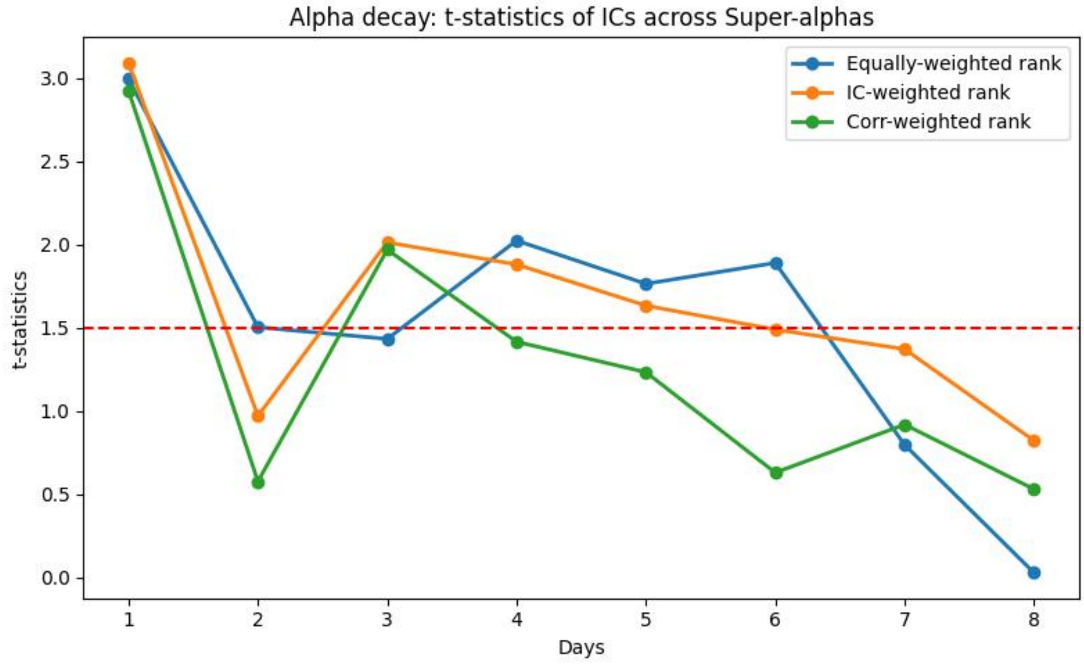

**2. Alpha decay duration.**

To measure how quickly the predictive power of an alpha diminishes, we compute the generalized information coefficient at horizon q:

$$IC(q)_t = corr(\alpha_t, r_{t+q}).$$

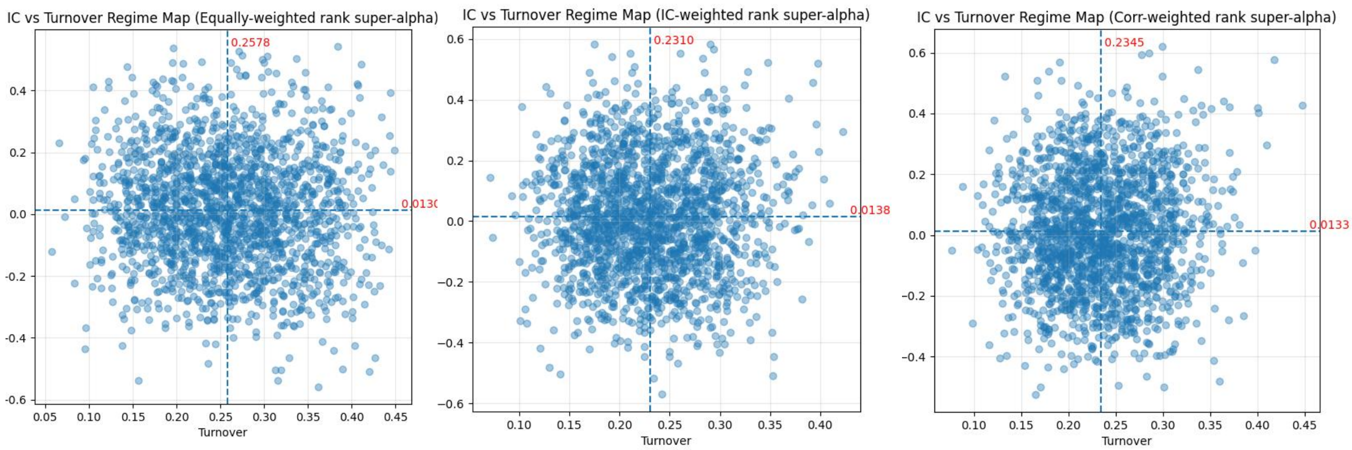
This metric captures how far into the future the signal remains informative. From the corresponding analysis, we observe that both IC-weighted and correlation-inverse-weighted super-alphas exhibit high-frequency characteristics, with decay duration of less than one day.

In contrast, the equally weighted super-alpha shows a slower decay, with a duration of approximately three days, indicating a more persistent signal. A plausible explanation is that weighting schemes emphasizing stronger or less correlated signals tend to concentrate on short-term, rapidly updating information, while equal weighting preserves more diverse components, including slower-moving effects.




This observation is also consistent with the average holding period of the underlying meta-alphas, which ranges from 0.6 to 6.4 days. Consequently, a daily re-balancing strategy is appropriate when implementing the portfolio in the next part.

**3. Turnover rate.**



Turnover measures the extent of portfolio re-balancing over time and is analyzed jointly with IC performance for each super-alpha. Our results indicate that all constructed super-alphas exhibit turnover rates of approximately 25%, which is reasonable within the context of cryptocurrency markets, where trading activity is typically higher and liquidity is relatively abundant.

## Portfolios constructions based on super-alphas:

In the final stage, we construct portfolios based on the derived super-alphas and evaluate their performance on both training and test datasets. Guided by the alpha decay analysis, we adopt a **daily re-balancing strategy**, as most signals exhibit short holding horizons.

To avoid look-ahead bias, we assume that alpha signals are computed immediately after market close using only information available up to that point. Positions are then established at the next day's open and held intraday until the close. Under this assumption, daily portfolio returns can be computed as a time series, from which performance metrics such as the annualized Sharpe ratio are derived. Given that cryptocurrency markets operate continuously (24/7), the annualization factor is adjusted accordingly:

$$sharpe\ ratio = \sqrt{365}\frac{\mathbb{E}[r]}{Std(r)}$$

, where r denotes daily returns. Furthermore, we choose the simple strategy of going long for all alive coins as a benchmark strategy and also include it into our analysis.

We highlight several key parameters in the portfolio construction:

· ***Short (Boolean):***

If set to *True*, the strategy takes both long positions in top-ranked assets and short positions in bottom-ranked assets based on the alpha signal. If set to *False*, the strategy is long-only. This distinction reflects practical constraints, as not all cryptocurrency markets support short selling.

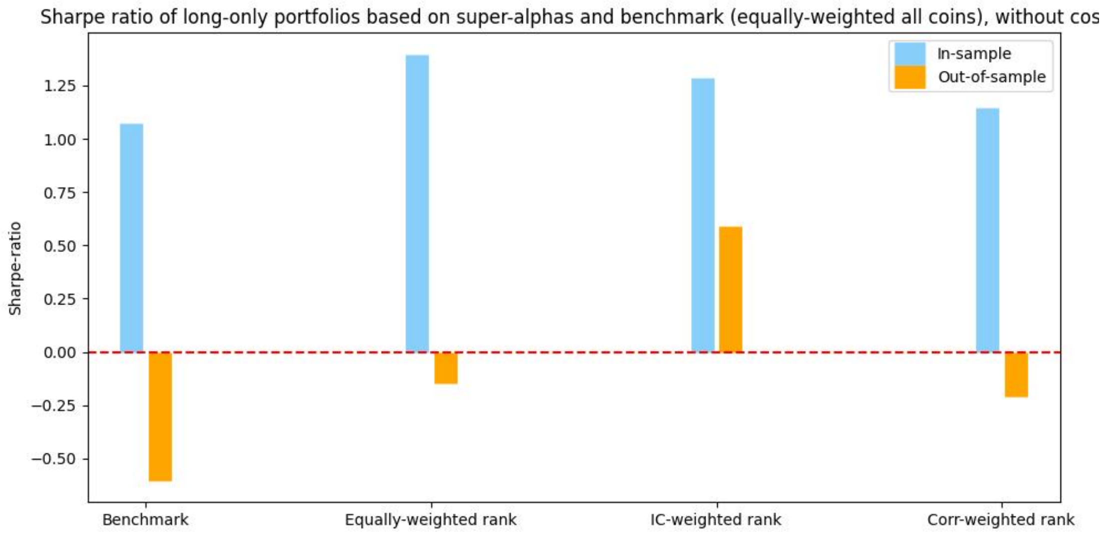
· ***Percentage (float, between 0 and 1):***

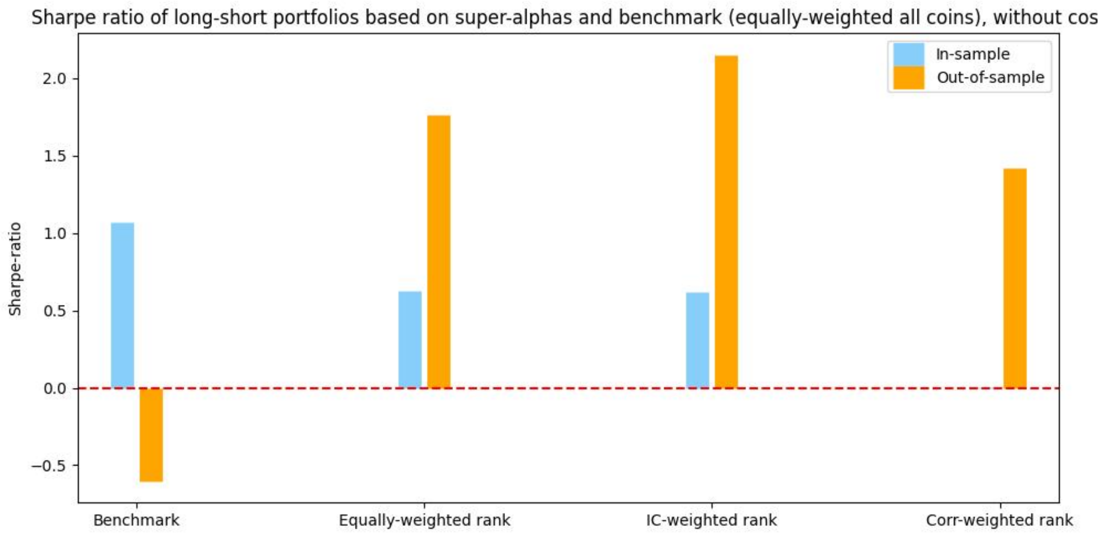
This parameter determines the fraction of assets selected for long and short positions. For example, if 40 assets are available and the percentage is set to 0.2, the top 20% (8 assets) are held long and the bottom 20% (8 assets) are held short. Specially, if ***percentage*** is set to 1 and ***short*** is set to False, it represents our benchmark strategy, whose results are irrelevant with the input alphas.

· ***Cost (float, typically between 0 and 0.1):***

This parameter represents transaction costs associated with portfolio rebalancing. It acts as a penalty proportional to turnover, capturing realistic trading frictions. Since cryptocurrency markets generally involve non-negligible transaction costs, incorporating this parameter improves the practical relevance of the backtesting results.

We first consider the non-cost case. The results are presented in the followings:




Some conclusions could be drawn in the above graphs:

· All super-alpha-based strategies outperform the benchmark in both in-sample and out-of-sample periods. It should be noted, however, that 2025 represents a relatively bearish phase for the cryptocurrency market. As a result, the Sharpe ratios of the equally weighted and correlation-inverse-weighted rank super-alpha portfolios are slightly negative out-of-sample, although they still outperform the benchmark. In contrast, the IC-weighted rank super-alpha portfolio delivers stable positive returns, achieving Sharpe ratios of approximately 1.25 in-sample and 0.5 out-of-sample. This suggests that weighting schemes incorporating predictive strength are more robust under adverse market conditions.

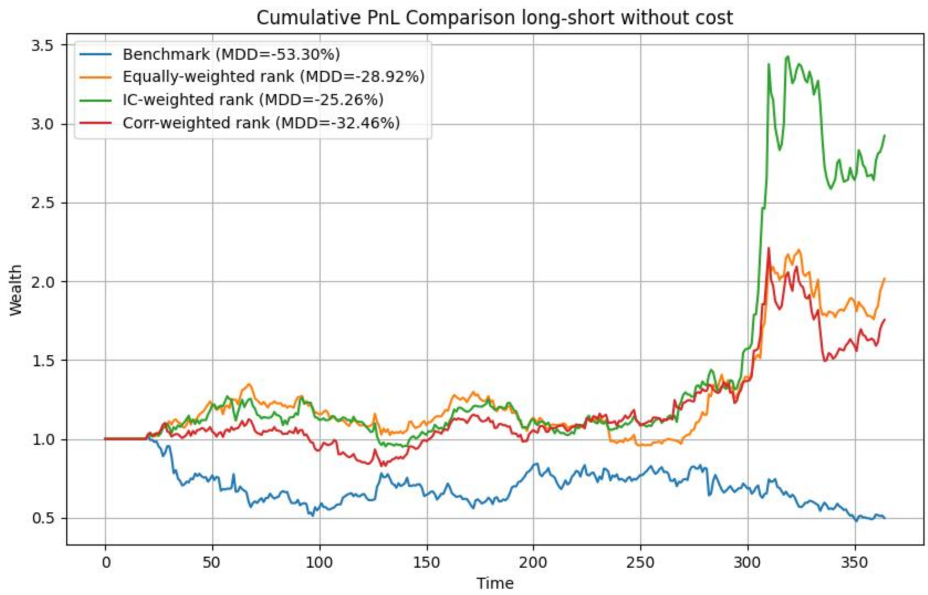
· Interestingly, super-alpha-based strategies underperform the benchmark in-sample. A possible explanation is that the in-sample period is dominated by strong market trends, during which long-only exposure benefits from overall upward momentum, while short positions detract from performance.

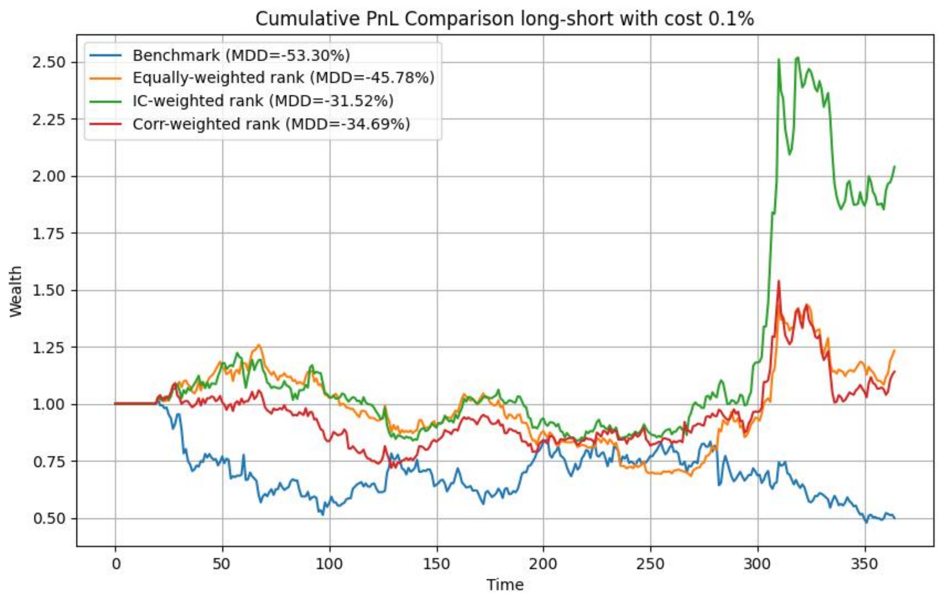
However, the out-of-sample results are particularly striking: the long-short strategies significantly outperform the benchmark, even during a bearish market regime. For example, the IC-weighted rank super-alpha portfolio achieves an annualized Sharpe ratio of approximately +2.1. This improvement can be attributed to the market-neutral nature of long-short strategies, which allows them to capture relative value opportunities and mitigate market direction risk. In addition, the diversification effect from combining multiple alphas becomes more pronounced out-of-sample, leading to stronger and more stable performance.

In the next step, we incorporate a transaction cost of 0.1% to evaluate the strategies under more realistic trading conditions.


As expected, the inclusion of transaction costs leads to a decline in the Sharpe ratios of all super-alpha-based strategies, while the benchmark strategy remains largely unaffected due to its negligible turnover. Encouragingly, the out-of-sample performance of the IC-weighted rank super-alpha remains robust: its Sharpe ratio is still approximately 1.5 after accounting for costs. This indicates that the strategy retains strong predictive power and is resilient to trading frictions, highlighting its practical viability in real-world implementation.

Finally, we compute and visualize the cumulative PnL of all strategies alongside the benchmark (assuming an initial capital of 1) on the test dataset. The results are presented both with and without transaction costs.




Encouragingly, all super-alpha-based portfolios outperform the benchmark, which is consistent with the previous analysis and demonstrates robustness under bearish market conditions as well as trading frictions.

In addition, we evaluate the risk profile of each strategy by computing the maximum drawdown (MDD). Most super-alpha-based portfolios exhibit MDDs in the range of -20% to -40%, which is generally acceptable in the cryptocurrency market and represents a clear improvement over the benchmark, which experiences a drawdown of approximately -50%. This suggests that the proposed strategies not only enhance returns but also provide better downside risk control.

## Conclusions and perspectives:

### Conclusion

This project investigates the applicability of the *WorldQuant 101 Alphas* framework in the cryptocurrency market and demonstrates that, although individual alphas often suffer from weak out-of-sample performance, their combination into well-constructed super-alphas can significantly improve robustness and predictive power.

Through careful alpha selection based on IC t-statistics, rank normalization, and diversified weighting schemes, we construct composite signals that exhibit stable performance across different market regimes. Empirical results show that these super-alpha-based portfolios consistently outperform a benchmark equal-weighted strategy in both in-sample and out-of-sample periods. Notably, the IC-weighted super-alpha achieves strong risk-adjusted returns, with Sharpe ratios remaining robust even after accounting for transaction costs.

From a risk perspective, the strategies demonstrate improved drawdown characteristics compared to the benchmark, highlighting the benefit of diversification across multiple signals. Overall, the findings confirm that combining multiple weak but complementary signals is an effective approach to building scalable and resilient quantitative strategies in the highly volatile cryptocurrency market.

### Perspectives

A particularly interesting direction for further research is to better understand the observed discrepancy between in-sample and out-of-sample performance of long-short strategies. Despite strong in-sample IC statistics, long-short portfolios underperform the benchmark in-sample while significantly outperforming out-of-sample. One plausible explanation is the influence of **market beta**: during strongly trending (bullish) periods, long-only exposure benefits from market direction, whereas long-short strategies may neutralize this effect.

To investigate this hypothesis, future work could involve regime-dependent backtesting, for example, applying long-only strategies during bullish markets and long-short strategies during bearish or sideways markets.

Another promising direction is to incorporate more realistic and sophisticated transaction cost models. In practice, trading costs are often nonlinear and depend on factors such as liquidity, market impact, and trade size. Extending the backtesting framework to include nonlinear cost structures would provide a more accurate assessment of strategy performance and implement ability.

More broadly, future research could explore dynamic weighting schemes, adaptive alpha selection, and machine learning methods to further enhance the stability and performance of super-alphas.